

7. A turntable is rotating at a constant number of revolutions per second. What is the relationship between the angular velocity of a point on the turntable and the distance of the point from the centre of the turntable?
- A Angular velocity is directly proportional to the distance.
 - B Angular velocity is inversely proportional to the distance.
 - C Angular velocity is directly proportional to the distance squared.
 - D Angular velocity is independent of distance.

Ans: D

Given number of revolutions per second, $f = \text{constant}$

$\therefore$ angular velocity $\omega = 2\pi f$ is a constant, independent of distance.

8. A satellite of mass M is in circular orbit with radius r around the Earth with a period of